Solving fermionic sign problems with PIKs

Kenji Fukushima,[1] Friederike Ihssen,[1] Renzo Kapust,[2] and Jan M. Pawlowski[2,3]

[1] *Theoretische Physik III, Ruhr-Universität Bochum, Universitätsstraße 150, 44721 Bochum, Germany*
[2] *Institut für Theoretische Physik, Universität Heidelberg, Philosophenweg 16, 69120 Heidelberg, Germany*
[3] *ExtreMe Matter Institute EMMI, GSI, Planckstr. 1, 64291 Darmstadt, Germany*

We extend the generative sampling approach with physics-informed kernels (PIKs) put forward in [1] to general physics systems with complex distributions, including fermionic systems. Central to this application is the resolution of sign problems of generic weighted sums of complex Gaußian distributions with the PIK-approach. This reduces the sampling costs of these systems to that of a real normal distributions. In a second step such a weighted sum distribution are evolved with a second PIK into the distribution of the theory at hand. In combination we reduce the sampling task for general complex systems to a Gaußian one, and the main computational task is that of computing the PIK. This task is at worst as expensive as standard HMC and we consider this a minor price to pay for solving sign problems.

I. INTRODUCTION

Lattice field theory provides access to many interesting strongly correlated physics systems and subject to efficient sampling architectures. Still, even qualitative access eludes us in systems with sign problems, ranging from, mostly fermionic, statistical and condensed matter systems to QCD and Beyond Standard Model physics. Moreover, realtime physics requires 'sampling' from a phase which constitutes a sign problem.

The sign problem is NP-hard and hence even the most efficient sampling algorithm with run out of steam eventually if one tries to overcome the problem by brute force. Still, efficiency goes a long way as in most cases is suffices to delay the onset of the sign problem or its scaling in the continuum limit. Apart from efficiency there is a reparameterisation of the sampling task such that the sign problem is absent. Both aspects have been tried in the past. Prominent examples for reparametrisation approaches to the sampling tasks are sampling with CLEs, the Leschetz thimble approach, and dual variables. In all these approaches, efficient and accurate sampling is also chiefly important. A prominent example for the direct combination of reparametrisation and brute force is sampling with the density of states. In short, practical solutions to the sign problem will most likely combine both aspects.

In [1, 2] a generative architecture for lattice field theories, sampling with Physics-Informed Kernels (PIKs), was put forward, which naturally encompasses both aspects in a unified picture. It is based on a Fokker-Planck type evolution or flow equation for the distribution $p(\varphi)$ of the theory under investigation. This flow equation, the Wegner equation [3], accommodates general reparametrisations and deformations of the distribution. The PIK is the differential map that encodes both the flow of the distribution and that of the sampling manifold. This formulation allows to map the sampling task in a given theory to that in a simple one, in most cases a real normal distribution. Moreover, it has veen shown in [1] that Wegner flows have a property, which is crucial for sampling theories with sign problems: they are *weight-preserving* for general PIKs. This means that real sampling is always mapped to real sampling (no sign problem) and the probability weight of all intervals is preserved (no reweighting problem). This entails in particular that if one can construct a PIK, that connect the sampling with a simple real positive distribution to that of a complex one, we have mapped the sampling of a sign problem to sampling a real simple distribution. It is this property which sets sampling with PIKs qualitatively apart from other approaches to the sign problem such as complex Langevin, Leschetz thimbles or the density of states.

The above immediately begs the question whether one can always construct a *real* PIK from some *real* distribution to a given complex one of a theory with a sign problem. In [1], a real PIK has been constructed for the simple realtime harmonic oscilator, hence solving the sampling task there within the PIK approach. The nature of the obstruction for constructing real PIKs, namely finding a sampling manifold on which the real part of the distribution does not vanish anywhere, indicates that this holds true for general realtime scalar theories, and by now this proof of principle with the harmonic oscillator has been extended to interacting scalar systems, [**?**].

In the present work we generalise the complex PIK approach to theories that do not allow for a direct application of real sampling and show that these theories can also be sampled from a real positive distribution, but the map to the physics disbtribution is done with a weighted sum of PIKs with positive relative weights, applied to a single real and simple distributions. Then, the weight-preserving property of the PIK and the positivity of the relative weights resolves the sign and potential overlap problem. This approach has a general applicability and completes the setup of the PIK-approach to sampling from complex distributions. We illustrate the properties of the setup at the example of a single site Hubbard model. This model is plagued with a sign problem for some chemical potential regimes, if sampled directly, but admits an analytic solution. We show, that the model can be sampled efficiently with the PIK approach in all regimes.

II. SOLVING SIGN PROBLEMS WITH PIKS

The PIK-approach [1, 2] reformulates the sampling problem of a normalised physics distribution $p(\varphi)$ with a sampling manifold $\mathcal{M}$ as a combination of two tasks: The sampling task is done with a simple $p_0(\phi)$, e.g. with a normal distribution, drawing samples from the sampling manifold $\mathcal{M}_0$. The second task is the (numerical) computation of the PIK $\dot\phi$ which is the differential map between the initial sampling pair $(p_0, \mathcal{M}_0)$ and the physics sampling pair $p_1, \mathcal{M}_1)$.

$$(p_0\,,\,\mathcal{M}_0) \xrightarrow{\dot\phi_t} (p_1\,,\,\mathcal{M}_1)\,, \qquad p_t = e^{-S_t}\,, \tag{1}$$

where the distribution p_t is normalised, $\int d\phi\, p_t(\phi) = 1$ for $t \in [0\,,\,1]$ and S_t is the likelihood of the distribution. Expectation values of operators $\mathcal{O}$ are then given by

$$\langle \mathcal{O}(\phi)\rangle_p = \frac{1}{N_s}\sum_n \mathcal{O}\left(\varphi(\phi_n)\right)\,, \tag{2}$$

where the ϕ_n are drawn from $\mathcal{M}_0$ with the probability p_0 and $\varphi(\phi)$ is the integrated PIK,

$$\varphi(\phi) = \phi + \int_0^1 ds\, \dot\phi(\phi_s)\,. \tag{3}$$

The PIK $\dot\phi$ is determined from the Wegner equation

$$\frac{d\,p_t}{dt} = -\frac{\partial}{\partial\phi}\left[\dot\phi\, p_t\right] \quad \leftrightarrow \quad \left(\frac{\partial}{\partial\phi} - \frac{\partial S_t}{\partial\phi}\right)\dot\phi = \frac{d\,S_t}{dt}\,, \tag{4}$$

whose solution has to be well-defined in a neigbourhood of the manifolds $\mathcal{M}_t$, embbeded in the complexification of $\mathcal{M}_0$. In its form for the distribution the generalised Fokker-Planck aspect of the Wegner equation is emphasised, and the embedding of other sampling approaches to the sign problem as specific PIKs has been discussed in [1].

The set of all manifolds $\mathcal{M}_t$ with $t = [0, 1]$ is called the PIKfold $\mathcal{P} = \{\mathcal{M}_t\}$ and a *real* PIKfold is defined by the sampling from the pair $(p_0\,, \mathcal{M}_0)$ is *real*. If a real PIKfold exists for a given physics problem with the complex distribution $p = p_1$, the respective complex sampling task is converted into a simple real one, see (2). The existence of a real PIKfold is in one-to-one correspondence to that of (4) in a neigbourhood of the sampling manifolds $\mathcal{M}_t$ for all $t \in [0, 1]$. A sufficient condition for the latter is [**JMP:** Is it?]

$$\mathrm{Re}\, p_t(\phi) \neq 0\,, \quad \text{for} \qquad \phi \in \mathcal{M}_t\,. \tag{5}$$

As already mentioned before, it is suggestive that scalar realtime QFTs satisfy (5), see [2**?**].

A. The PIKfan

In order to set up a general sampling architecture for systems with sign problem we have to also accommodate those that do not admit any real PIKfold but only complex ones. We shall argue that generic fermionic systems with a finite chemical potential fall into this category and we also show that quite generically theories without real PIKfolds suffers from an inherent initial sign problem: all possible sampling pairs $(p_0(\phi)\,, \mathcal{M}_0)$ with a complex distribution p_0 and a complex sampling manifold $\phi \in \mathcal{M}_0$ that are related to the physics sampling pair $p(\varphi)\,, \mathcal{M})$ have a sign problem themselves.

We start our analysis by using bosonisation to reformulate fermionic systems with sign problems as complex bosonic ones with the (non-local) action $S(\varphi)$. They define the distribution $p_1(\phi)$ as

$$p_1(\phi) = p(\phi), \qquad p(\varphi) = e^{-S(\varphi)}\,. \tag{6}$$

and $S(\varphi)$ does not satisfy (5) for *any* $\mathcal{M}_1$. Then, the system probably does not admit a real PIKfold.

Let us now assume that $p(\varphi)$ on some sampling manifold $\mathcal{M}_1$ is connected with a PIK to an appropriately chosen weighted sum of simple complex distributions with some sampling manifold $\mathcal{M}_0$. The simple complex distributions are typically Gaußian ones, ρ_i and we yield

$$p_0(\phi) = p^{(w)}(\phi)\,, \tag{7a}$$

with

$$p^{(w)}(\phi) = \sum_{i=1}^N w_i\,\rho_i(\phi)\,, \qquad \sum_{i=1}^N w_i = 1\,. \tag{7b}$$

where the $w_i \in \mathbb{R}^+$ are the relative weights of the corresponding *complex* 'normal' distributions ρ_i. We may also consider negative weights but this is not consistent with positive densities in the system and hence we defer the discussion of the more general case to future work. We shall discuss whether one can always find such a weighted sum at the end of this section.

Moreover, (7) may not admit a real initial sampling manifold $\mathcal{M}_0$ as (5) does not hold true for general weighted sums even in zero dimensions. Importantly, weighted Gaußian sums *always* admit a complex sampling manifold on which (4) is well-defined. Moreover, one can derive sampling pairs with weighted sum, that admit a complex PIKfold (1) with the physics distribution (6) as the final distribution. This derivation resorts to standard non-linear optimisation procedures. We emphasise that this part of the architecture does not require high precision numerics which makes it tailor mode for Machine Learning.

In summary we are led to a PIKfold

$$(p^{(w)}\,, \mathcal{M}_0) \xrightarrow{\dot\phi_t^{(w)}} (p_1\,, \mathcal{M}_1)\,. \tag{8}$$

Note also that typically the respective sampling pairs $p^{(W)}(\phi)$ in (7) have sign problems themselves if the final one has a sign problem. However, the sign problem

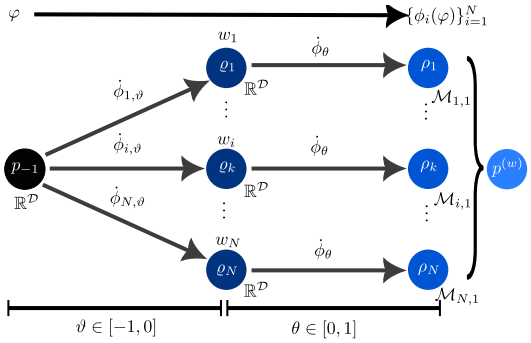


FIG. 1: Density (34) as a function of chemical potential μ for couplings $U = 1$ and $\beta = 10$. [**JMP:** Please use $\vartheta, \theta \to t$.]

of weighted sums can be solved with PIKs, and in the present work we suggest to use a fan of PIKs:

We choose a *real normal* distribution $p_{-1}(\phi)$ and the real sampling manifold $\mathcal{M}_{-1}$ as our original sampling pair. The sampling of the physics distribution is finally done with configurations sampled from p_{-1}. Now we use that any of the summand distributions ρ_i in (7) admits a *real* PIKfold with p_{-1},

$$(p_{-1}\,,\,\mathcal{M}_{-1}) \xrightarrow{\dot\phi_{i,t}} (\rho_i\,,\,\mathcal{M}_{i,0})\,, \tag{9}$$

see Figure 1. With this fan of PIKs we can map the complex sampling problem of the weighted sum of real ones with the *real* weights w_i. Note that this construction avoids the so-called residual sign problem present in the Lefschetz thimble approach. There, each thimble comes with a complex phase which would translate into a complex 'weight' w_i. Moreover, the number of contributing thimbles has a qualitatively larger growth rate than the number of distributions ρ_i. This is but one of the qualitative advantages of the present approach. In any case, this fan of PIKs solves the sign problem of $p_0(\phi) = p^{(w)}(\phi)$. Expectation values of operators $\mathcal{O}$ with the distribution p_0 can be written as

$$\int\limits_{\mathcal{M}_0} d\mu\, p^{(w)}(\phi)\, \mathcal{O}(\phi) = \sum_{i=1}^N w_i \int\limits_{\mathcal{M}_{-1}} d\mu\, \mathcal{O}\big(\phi_{i,0}(\phi)\big)\,. \tag{10a}$$

where

$$\phi_{i,t}(\phi) = \phi + \int_{-1}^t ds\, \dot\phi_{i,s}(\phi_s)\,, \tag{10b}$$

where the single maps all come with the real PIKfolds (9) and $t \in [-1, 0)$.

The second step from $t = 0$ to $t = 1$ in Figure 1 uses the PIK $\dot\phi_t^{(w)}$ defined in (8) for $t \in [0, 1)$. In combination this leads us to a fan of PIKs $\phi_{i,t}^{(w)}$ with $t \in [-1, 1)$, defined by

$$\dot\phi_{i,t}^{(w)} = \dot\phi_{i,t}\,\theta(-t) + \dot\phi_t^{(w)}\,\theta(t)\,, \tag{11}$$

with the Heaviside θ-function. This readily leads us to our final sampling prescription

$$\int\limits_{\mathcal{M}} d\mu\, p(\phi)\, \mathcal{O}(\phi) = \sum_{i=1}^N w_i \int\limits_{\mathcal{M}_{-1}} d\mu\, \mathcal{O}\big(\varphi_i^{(w)}(\phi)\big)\,. \tag{12a}$$

where

$$\phi_{i,t}^{(w)}(\phi) = \phi + \int_{-1}^t ds\, \dot\phi_{i,s}^{(w)}(\phi_s)\,, \quad \varphi_i^{(w)}(\phi) = \phi_{i,1}^{(w)}(\phi)\,, \tag{12b}$$

for $t \in [-1, 1)$. This completes our construction. Note that the combined PIKs (11) define non-trivial actions $S_{i,1}(w)$ for each i. Moreover, the $\dot\phi_i^{(w)}$ define Wegner flows from ρ_{-1} to $\rho_{i,1}^{(w)}$ with the sampling manifolds $\mathcal{M}_{i,1}$ and due to the weight-preserving property of the PIKs, see [1], each of the sampling pairs $(\rho_{i,1}^{(w)}, \mathcal{M}_{i,1})$ is real. In short, we have converted the task of sampling the complex distribution $p(\varphi)$ that does not admit a real PIKfold, into the task of sampling from a sum of real sampling pairs, hence solving the sign problem.

The practical sampling procedure is done by drawing a set of configurations $\mathcal{S}_{-1} = \{\phi_n\}$ with the distribution ϕ_{-1}. The expectation values of operators $\mathcal{O}$ with the physics distribution $p(\varphi)$ is then provided by

$$\langle \mathcal{O}(\phi)\rangle_p = \frac{1}{N_s} \sum_n w_i\, \mathcal{O}\left(\varphi_i^{(w)}(\phi_n)\right)\,, \tag{13}$$

over all N_s is the number of configurations in $\mathcal{S}_{-1}$.

B. Finding PIKfans

This setup now leaves us with the task of finding the PIKfan $\{\dot\phi_{i,t}\}_{i=1}^N$ that cummulatively maps p_{-1} to a modelled distribution

$$\tilde p(\phi) = \sum_{i=1}^N w_i\, \rho_{i,1}^{(w)}(\phi)\,, \tag{14}$$

that approximates the targeted distribution p well. Each rib $\dot\phi_{i,t}$ of the PIKfan can be expanded in a basis whose free parameters we denote by k_i. As was shown explicitly in [1], when mapping p_{-1} to a complex Gaußian, k_i can be boiled down to the respective covariance and mean value of the Gaußian. Including the weight w_i for the i-th rib of the PIKfan, it encompasses the free parameters $\Theta_i := \{k_i, w_i\}$. Thus we can cast the task of finding PIKfans into jointly optimising the parameters $\Theta := \{\Theta_i\}_{i=1}^N$.

At first sight, this may look like a straight-forward optimisation problem that can be tackled by minimising known divergences like the Kullback–Leibler or Fisher divergence. However, note that, firstly, naïvely computing said divergences itself generally suffers from a sign problem as it will be given as a high-dimensional integral w.r.t. a complex distribution. Secondly, even if one

would be able to compute these divergences efficiently, as both p and $\tilde{p}$ are complex, the divergence will take up complex values, conceptually hindering standard optimisation methods like gradient descent. Here, we will solve both problems constructively by using the weight-preserving property of the PIKs.

In a nutshell, we are interested in writing down a divergence $D_g(\tilde{p}\|p)$ with an appropriately chosen function $g : \mathbb{C}^2 \to \mathbb{R}$ that vanishes if and only if $\tilde{p} = p$. Furthermore, $D_g(\tilde{p}\|p)$ should be computable without suffering from a sign problem and itself take up values in $\mathbb{R}$ in order to allow for standard optimisation methods of the parameters Θ.

To that end, we define the divergence $D_g(\tilde{p}\|p)$ as

$$D_g(\tilde{p}\|p) = \sum_{i=1}^{N} w_i \int_{\mathcal{M}_{i,1}} D\phi\, \rho_{i,1}^{(w)}(\phi)\, g[\,\tilde{p}(\phi), p(\phi)\,]\,, \quad (15)$$

For this definition, we can use that while the mixture of the complex distributions $\rho_{i,1}^{(w)}$ posses a sign problem, it is a sign problem that we easily treat by using (10) and (12). Using this property, we can rewrite the above divergence as a high-dimensional integral w.r.t. the real distribution p_{-1} on the initial real manifold $\mathcal{M}_{-1}$ via

$$D_g(\tilde{p}\|p) = \sum_{i=1}^{N} w_i \int_{\mathcal{M}_{-1}} D\phi\, p_{-1}(\phi)\, g[\,\tilde{p}(\varphi_i^{(w)}(\phi)), p(\varphi_i^{(w)}(\phi))\,] \quad (16)$$

[**JMP:** Here we suggest the two procedures with the KL divergence or the Fisher metric or by minimising the PIK from $p^{(w)}(\phi)$ to $p(\varphi)$.]

III. ONE-SITE HUBBARD MODEL

A. Model definition and analytical solution

The hubbard model suffers the sign problem in the same way as QCD at finite density. Since it describes the lattice fermions, the model has a conserved charge corresponding to the fermion number, which can be coupled to the chemical potential μ. In the tight-binding limit, fermions can no longer hop to neighboring sites, and the model Hamiltonian is decomposed into the one-site Hamiltonian at each site. More specifically, using the number operator for up and down spin fermions, $\hat{n}_\uparrow$ and $\hat{n}_\downarrow$, the one-site Hamiltonian reads:

$$\hat{H} = U\hat{n}_\uparrow\hat{n}_\downarrow - \mu(\hat{n}_\uparrow + \hat{n}_\downarrow)\,. \quad (17)$$

This Hamiltonian is simple enough to be analytically solvable, and yet, it still captures the essence of the sign problem.

In the operator formalism, it is a straightforward task to compute the partition function using the Fock state basis. Because $\hat{n}_{\uparrow,\downarrow}$ takes an eigenvalue of either 0 or 1, there are four eigenvalues of $\hat{H}$, namely, 0, $-\mu$, $-\mu$, $U - 2\mu$. Therefore, the partition function is

$$Z = \mathrm{tr}\big[e^{-\beta\hat{H}}\big] = 1 + 2\,e^{\beta\mu} + e^{\beta(2\mu-U)}\,, \quad (18)$$

from which we can compute thermodynamic quantities.

The equivalent representation in the path-integral formalism is

$$Z = \mathcal{N}\int d\phi \left(1 + e^{\beta(i\phi+\mu+U/2)}\right)^2 e^{\beta\phi^2/2U} \quad (19)$$

with the normalization, $\mathcal{N} = \sqrt{\beta/(2\pi U)}$. The quantity inside the parentheses is a counterpart of the Dirac determinant in QCD at finite density. We emphasize that the similarity is attributed to the fermionic nature and in this sense the one-site Hubbard model can be regarded as a well-motivated toy model to investigate the sign problem in QCD at finite density.

Indeed, if we consider the strong-coupling limit of QCD in the hopping parameter expansion, the leading-order Dirac determinant reads:

$$\sim \prod_{\text{phase space}} \det\big(1 + L\,e^{\beta\mu}\big)\big(1 + L^\dagger\,e^{-\beta\mu}\big)\,, \quad (20)$$

where L is an $SU(3)$ element corresponding to $e^{\beta i\phi} \in U(1)$ in the one-site Hubbard model. We observe that, in the same way as the tight-binding Hubbard model, the spatial hopping is suppressed, and the above determinant is a simple product of the quantity evaluated at each phase-space point. Then, it is easy to see the origin of the sign problem. For simplicity, let us keep terms up to the linear order in terms of L, and then, we can expand the determinant as

$$\sim 1 + \mathrm{tr}\,L\,e^{\beta\mu} + \mathrm{tr}\,L^\dagger\,e^{-\beta\mu}\,. \quad (21)$$

at each phase-space point. Obviously, for $\mu \neq 0$, the Hermiticity is broken, and the eigenvalue takes a complex value which may have a negative value in the real part.

Coming back to Eq. (19), it is useful to split the path integral as $Z = Z_0 + Z_1 + Z_2$, where

$$Z_0 = \mathcal{N}\int d\phi\, e^{\beta\phi^2/2U}\,, \quad (22a)$$

$$Z_1 = 2\mathcal{N}\int d\phi\, e^{\beta\phi^2/2U + i\beta\phi + \beta(\mu+U/2)}\,, \quad (22b)$$

$$Z_2 = \mathcal{N}\int d\phi\, e^{\beta\phi^2/2U + 2i\beta\phi + 2\beta(\mu+U/2)}\,. \quad (22c)$$

These are Gaussian integrals individually. The first integral of Z_0 is 1 with a proper choice of the normalization $\mathcal{N}$. The next one, Z_1, looks complex, but noting that $\beta\phi^2/2U + i\beta\phi + \beta(\mu + U/2) = \beta(\phi + iU)^2/2U + \beta\mu$, we can make a variable shift as $\phi + iU \to \phi$ to find that the Gaussian integral leads to $2e^{\beta\mu}$, that is nothing but the second term in Eq. (18). We can shift $\phi + 2iU \to \phi$ for Z_2 to reproduce the third term in Eq. (18).

One might think that the one-site Hubbard model is too simple, but once the fermionic determinant is included in the action, we see that the total action is a good proxy for the QCD sign problem, that is,

$$S(\phi) = \frac{\beta}{2U}\phi^2 - 2\ln\left[1 + e^{\beta(i\phi+\mu+U/2)}\right] . \tag{23}$$

We point out that this action has some characteristic points. First, we can easily identify the zeros of the partition function where the action is singular. These singular points are aligned along

$$\phi_{\mathrm{s},n} = (2n+1)\frac{\pi}{\beta} + i\left(\mu + \frac{U}{2}\right) . \tag{24}$$

Second, we can find the saddle points (sometimes called the critical points) from the condition, $S'(\phi) = dS(\phi)/d\phi = 0$, which are denoted by $\phi_{\mathrm{c},n}$. There is no simple analytical form for $\phi_{\mathrm{c},n}$. However, we can still find approximate locations depending on the parameter regions of $\mu+U/2$ in the action if $\beta U \ll 1$ holds. Specifically, for the sign problem, the most nontrivial alignment of the saddle points realizes as

$$\phi_{\mathrm{c},n} \simeq \frac{2\pi n}{\beta} + i\left(\mu + \frac{U}{2}\right) - \frac{i}{\beta}\ln\left[\frac{\mu + U/2}{2U - (\mu + U/2)}\right] \tag{25}$$

for $0 \lesssim \mu + U/2 \lesssim 2U$ where the sign problem becomes severe. If we choose the model parameter such that $\beta/U \ll 1$, as is the case in later discussions, the imaginary part is dominated by the first term only, and the approximate saddles points should finally become

$$\phi_{\mathrm{c},n} \simeq \frac{2\pi n}{\beta} + i\left(\mu + \frac{U}{2}\right) . \tag{26}$$

B. Structures of the Lefschetz Thimbles

Before the PIKS application to the one-site Hubbard model, it would be instructive to examine the structures of the Lefschetz thimbles in this model defined by the action (23). In this one-variable example, the Lefschetz thimbles are simply the steepest descents starting from the saddle points. For numerical results, we will fix $U = 1$ and $\beta = 10$ throughout this paper. Then, in the previous analysis, it is already known that the sign problem is severe in the density region, $-1/2 \lesssim \mu \lesssim 3/2$.

IV. APPLICATIONS

The present section begins with a naive application of the PIK to the one-site Hubbard model in Section IV A, outlining its failure when approaching the poles described in (24).

This is followed by two applications of the weighted sum procedure proposed in Section II. In Section IV B, we begin by discussing an exact split of the one-site Hubbard model action into Gaussians, as was alluded to in (22a). In this case, the second step in the two-step in Figure 1 becomes trivial, as the physical action is already given by the action at $t = 0$. While this split is always possible in fermionic theories, the increasing number of contributing Gaussians to the exact split quickly gets out of hand with increasing system size.

Thus, in Section IV C we consider a second example, which uses a slightly less informed ansatz that uses a complex sampling manifold at $t = 0$, which can be evolved to actions with $\mu \leq 0$.

A. Poles in the PIKfold

In distinction to the examples considered in [1], fermionic systems introduce a new problem for the PIK approach: the presence of poles in the action. We find, that whenever the PIKfold approaches one of the poles within the action, a cut starts to develop in the manifold. Due to the layer-wise architecture of the PIK-approach and the full analytic access to each layer, this observation is easily explained: A pole in the target action S_t, directly translates into a pole of $\dot{\phi}_t$, by virtue of solving the Wegner equation (4). Recalling (C6), $\dot{\phi}(\phi)$ determines the change of a given sample ϕ. Thus, samples in the vicinity of the pole are either attracted or repulsed by it. In particular the latter case initiates a cut right at the position of the pole, which does not close generically at later RG-times: The PIKfold has ripped!

While in principle, the ripping of the PIKfold is not an issue for the computation itself, a ripped manifold is no longer a valid integration contour and can not be used to sample observables.

To demonstrate the impact of poles on the PIKfold we begin by considering a simple path which evolves samples from a Gaussian distribution at $t = 0$ to the full action (23) at $t = 1$

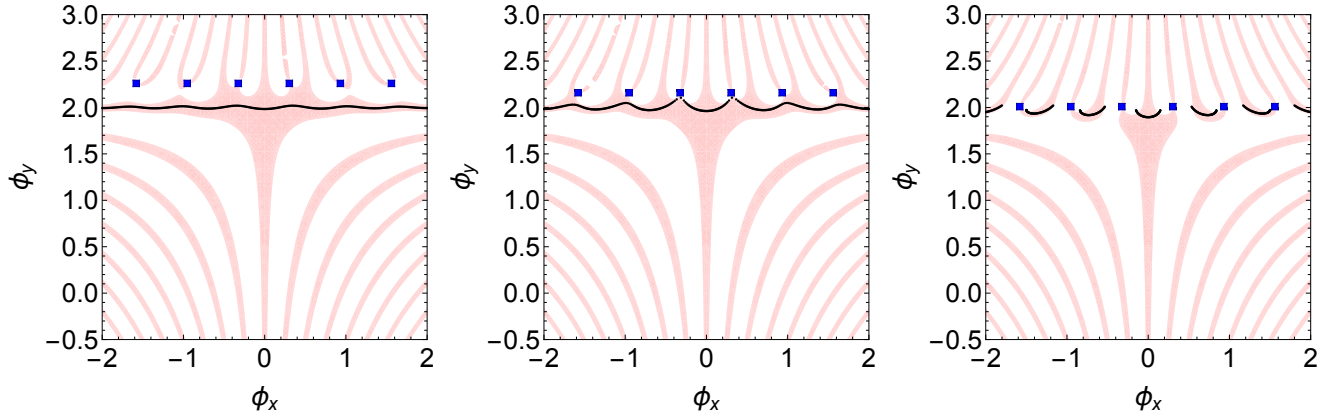
$$S_t(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{t\beta\left(i\phi+\mu+\frac{U}{2}\right)}\right] + \mathcal{N}_t . \tag{27a}$$

with time dependent poles

$$\phi_0 = (2n+1)\frac{\pi t}{\beta} + i\left(\mu + \frac{U}{2}\right) . \tag{27b}$$

The path (27a) allows to determine a analytical expression for $\dot{\phi}$ that solves the Wegner equation (4). The normalisation $\mathcal{N}_t$ is fixed by the constraint $|\dot{\phi}(\phi)| \leq \infty$ as $\phi \to \infty$. The PIKfold at $t = 0$ is initiated on the real axis $\mathcal{M}_0 = \mathbb{R}$, where Gaussian samples can be drawn. These are then evolved using (C6) and the corresponding $\dot{\phi}$ for the path (27a).

Throughout the RG-time evolution of the distribution, the poles remain at a constant imaginary value and only move in the direction of the real axis. Hence, any PIKfold



(a) $\mu = 1.75$ (b) $\mu = 1.65$ (c) $\mu = 1.5$

FIG. 2: Final sampling manifold for the single-PIKfold approach discussed in Section IV A for different densities. The sampling manifold is indicated in *black*, whereas the *blue* dots indicate the poles in the action. The red areas are inserted for orientation and indicate $\mathrm{Im}\, S(\phi) \pmod{2\pi} < \frac{\pi}{4}$ or $\mathrm{Im}\, S(\phi) \pmod{2\pi} > \frac{7\pi}{4}$.

that remains below $\mathrm{Im}[\mathcal{M}_t] \lesssim \mu + \frac{U}{2}$ does not encounter the poles.

We depict the final sampling manifold for this example in Figure 2 for the parameter choices $U = 1$, $\beta = 10$ and various densities. The results displayed in figures (a) and (b) correspond to $\mu = 1.75$ and $\mu = 1.65$ and are valid sampling manifolds as they remain far away enough from the poles and show no ripping. In (c) at $\mu = 1.5$, the poles lie on the sampling manifold. As such it does no longer form a closed integration contour and we can no longer evaluate observables from the samples.

The complex structure of the one-site Hubbard model depicted in Figure 2 allows to make the following observation: For densities $\mu \gtrsim 1.5$ there exists a contour with relatively constant phase at $\mathrm{Im}[\phi] \approx 2$, which is broken once the poles pass trough it, i.e. $\mu \lesssim 1.5$. This observation allows to distinguish a first type of sign problem in the one-site Hubbard model: All actions with $\mu \gtrsim 1.5$ are connected to sampling from a real Gaussian by means of a single real PIKfold. The solution of the PIKfold itself looks very similar to the expression obtained for a single complex Gaussian, which is explained by the fact, that the action (18) is dominated by the third contribution in (22a).

[**FI:** I think these are also the solutions that can be obtained from a single thimble] [**FI:** I put some remnant results in the appendix. Not sure what else we want to have]

B. Extension to weighted sums

As stated in the previous sections, in cases with a particularly severe sign-problem $\mu \lesssim 1.5$ a real PIKfold fails to provide a closed contour/manifold, i.e. it rips, as is illustrated in Figure 2c.

For such cases, we make use of the PIKfan: Due to our analytic grasp of the one-site Hubbard model we can disentangle the action and rewrite it as a sum of three Gaussian distributions (7) with

$$\begin{aligned} \rho_{1,t}(\phi) &= e^{-\frac{\beta}{2U}\phi^2} & w_{1,1} &= 1/\mathcal{N}_1 \\ \rho_{2,t}(\phi) &= e^{t\beta\left(\mu+\frac{U}{2}+i\phi\right)-\frac{\beta}{2U}\phi^2} & w_{2,1} &= 2/\mathcal{N}_1 \\ \rho_{3,t}(\phi) &= e^{2t\beta\left(\mu+\frac{U}{2}+i\phi\right)-\frac{\beta}{2U}\phi^2} & w_{3,1} &= 1/\mathcal{N}_1\,. \end{aligned} \tag{28}$$

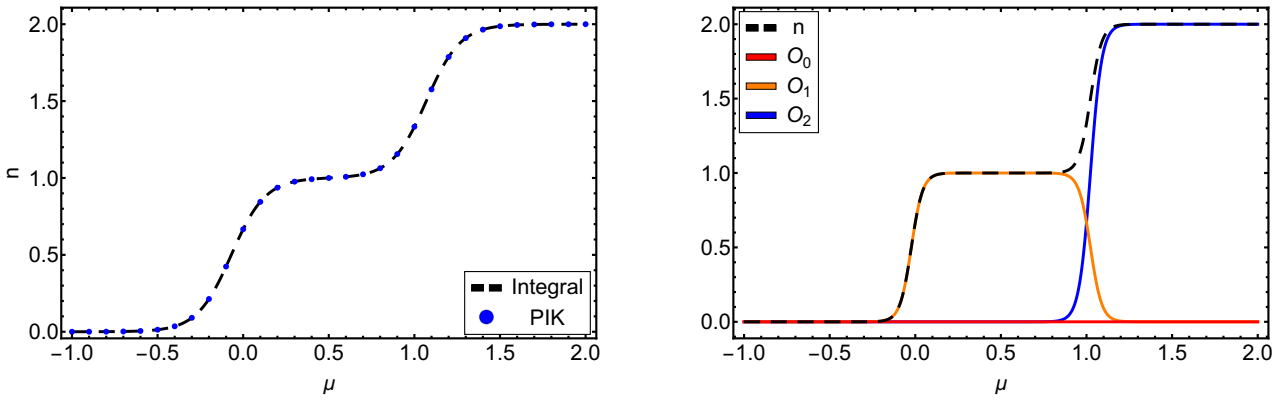
where $\mathcal{N}$ is the overall normalisation at $t = 1$ and $\sum_i w_i \rho_i = p(\phi)$ is the full action of the one-site Hubbard model. Note, that this ansatz is also possible for larger lattices in the Hubbard model. However, the number of required Gaussians scales like the fermion determinant with the lattice volume.

In the present context we consider is nevertheless as an instructive example of a trivial PIKfan: Each of these contributions is a (shifted) Gaussian and allows for the analytical computation of a sign-problem free contour using a real PIKfold. This corresponds to the first step in Figure 1. [**FI:** I am here...]

Instead of using a single complex PIK, we proceed to add three, corresponding to the respective summands in (28). The first term remains as is: We sample from the real Gaussian. The second and third term are described by an analytically given PIK, which solves the equation

$$\beta\left(\mu + \frac{U}{2} + i\phi\right) + \partial_t n = \partial_\phi \dot{\phi} - \dot{\phi}\left(it\beta - \frac{\beta}{U}\phi\right)\,, \tag{29}$$

where t is integrated up to $t = 1$ and $t = 2$ for the second and third term respectively. This ODE has a general analytic solution, which is only finite for all ϕ if



(a) Result from the PIKfold compared to the exact solution at $\beta = 10$.

(b) Contributions from the different distributions (28) at $\beta = 30$.

FIG. 3: Density (34) as a function of chemical potential μ for couplings $U = 1$.

the normalisation is chosen as

$$\partial_t n = -\beta\left(\mu + \frac{U}{2} - tU\right) ,$$
$$\Rightarrow n_t = -\beta\left(\mu t + \frac{U}{2}t(1-t)\right) + n_0 , \qquad (30)$$

where n_0 is the normalisation of the original Gaussian at $t = 0$. With this the solution to the PIK follows as

$$\dot{\phi} = iU \quad \Rightarrow \quad \phi_t(\varphi) = \varphi + iUt , \qquad (31)$$

where φ is given by the original real field $\varphi = \phi_0 \in \mathbb{R}$ and

$$\varphi \sim e^{-\frac{\beta}{2U}\varphi^2} . \qquad (32)$$

By weighting the total distribution with the adjusted normalisations we can now sample general, holomorphic operators $O(\phi)$ with

$$\langle \hat{O} \rangle = \sum_i \frac{O(\varphi_i) + 2N_1 O(\varphi_i + iU) + N_2 O(\varphi_i + 2iU)}{1 + 2N_1 + N_2} , \qquad (33)$$

where $N_t = e^{n_t}$ and we sum over Gaussian samples φ_i. To verify the calculation we sample the density operator, which is given by

$$n = -i\langle\phi\rangle . \qquad (34)$$

The μ-dependence of the density at $\beta = 10$ and $U = 1$ is displayed in Figure 3. We also show the solution of a direct evaluation of the integral as a comparison.

C. Complex initial sampling manifolds

Sometimes, it is also possible to sample from a complex distribution. For example, lets consider

$$S_\mu(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + e^{\beta\left(i\phi + \mu + \frac{U}{2}\right)}\right] . \qquad (35)$$

with the path parameter μ. At $\mu = 0$ we can use a decompositions into Gaussians for sampling

$$p_{\mu=0} = \frac{\mathcal{N}(0, \beta/U) + 2\mathcal{N}(iU, \beta/U) + e^{\beta U}\mathcal{N}(2iU, \beta/U)}{3 + e^{\beta U}} . \qquad (36)$$

To sample from the complex Gaussians, we can use a standard complex PIK. If we do that, we obtain a starting manifold $\mathcal{M}_0 = \mathbb{R} + \frac{2}{3}i$. This we can use to evolve our PIK along μ. The PIKfold for $\mu \in [-1, 2]$ is displayed in Figure 5a. While the PIKfold is well defined for $\mu < 0$, we find ripping at $\mu \approx 0.1$. This is shown also in Figure 4.

V. GENERALISED APPROACHES WITH ML

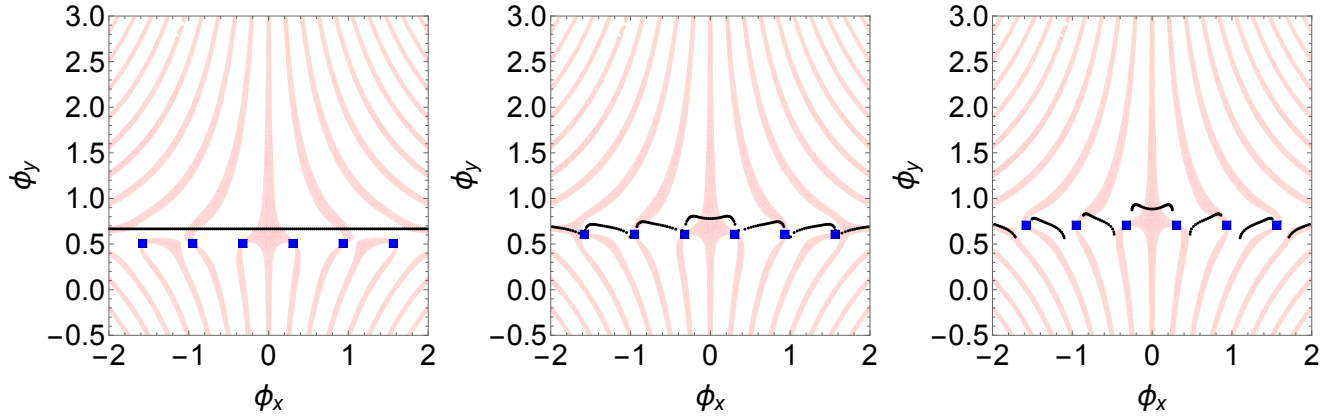
[**FI:** Finally Renzo presents his ML based general procedure!]

VI. RESOLVING THE PIK PAIR

Here we discuss constructive solutions for the determination of the PIKfold, or the PIKsubfolds IV B, specifically in the case of higher-dimensional fermionic theories. To begin with, we note that the distribution of many theories resembles closely weighted sums such as (7). This certainly holds true for the one-site Hubbard model but extends beyond this very restrictive example. Moreover, many multimodal distributions can be expanded about a weighted sum of normal distributions that are peaked at the peaks of the full distributions.

Moreover, in terms of PIKfolds these weighted sums can be collected into equivalence classes: two distributions $p(\phi)$ and $q(\phi)$ fall into the same equivalence class if they can be connected with a PIKfold $\mathcal{P} = \{\mathcal{M}_t\}$,

$$p \simeq q : \quad \exists\ (p_t\,, \mathcal{P}) \quad \text{with} \quad p_0 = p\,, \quad p_1 = q\,. \qquad (37)$$



(a) $\mu = 0.0$ (b) $\mu = 0.1$ (c) $\mu = 0.2$

FIG. 4: Sampling manifold for the single-PIKfold approach discussed in **??** for different densities. The sampling manifold is indicated in *black*, whereas the *blue* dots indicate the poles in the action. The red areas are inserted for orientation and indicate $\mathrm{Im}\, S(\phi) \pmod{2\pi} < \frac{\pi}{4}$ or $\mathrm{Im}\, S(\phi) \pmod{2\pi} > \frac{7\pi}{4}$.

Note that the equivalence (37) may be hard to prove as we can start with any sampling manifold $\mathcal{M}_0$. In the cases discussed below we start with the original sampling manifold of the underlying field theory, in our example theory, the one-site Hubbard model this is $\mathcal{M}_0 = \mathbb{R}$.

The key idea of the constructive solution of the sampling problem is the following.

(1) Finding p_t, weighted sum (7) or similar, as with $p_1 \simeq e^{-S}$.

(2) Sampling from p_t.

Both steps are sufficiently generic to allow for different solving strategies best-suited for specific problems. Below we shall discuss several ones for (1) and a tailor made one for (2). These strategies are illustrated with specific ones at the example of the one-site Hubbard model. Their applications to fermionic quantum field theories beyond the analytic example of the one-site Hubbard model goes beyond the scope of the present more conceptual work and will be presented in a forthcoming work.

A. Resolving the complex structure

In this Section we work out and test a two step approach to the PIK-sampling of complex distributions in the presence of ripping. In the first step we resolve the ripping with the PIK-pair

$$(\bar{S}_t\,,\,\bar{\mathcal{M}}_t)\,, \tag{38}$$

without ripping for all $t \in [0,1]$ and $(\bar{\rho}_0\,,\,\bar{\mathcal{M}}_0)$ being real for the sake of simplicity. Other choices of the initial sampling pair are possible and do not modify the procedure. Evidently such a choice excludes $\bar{S}_1 = S$ if the latter does not allow a PIKfold. Accordingly we have

$$\bar{S}_t = S'_t + \mathcal{T}_t\,, \qquad S'_0 \equiv 0\,, \quad S'_1 = S\,. \tag{39}$$

with $\bar{p}_0 = \exp\{-\mathcal{T}_0\}$ is the initial simple distribution and $\mathcal{T}_t$ prevents the ripping of the manifold $\mathcal{M}_t$. Specifically, T_1 arranges for $\mathrm{Re}(\bar{p}_1) \neq 0$ on $\mathcal{M}_1$. We know from [1] that the set of $\mathcal{T}_1$ can be computed from S alone: all admissible $\mathcal{T}_t$ can be inferred from the condition $\mathrm{Re}(\bar{p}_1) \neq 0$ on $\bar{\mathcal{M}}_1$. However, in a high-dimensional data space this entails the search of critical points in a high dimensional space, and in general the computational complexity of this task rivals that of solving sign problems. Moreover, we also need a trajectory $\mathcal{T}_t$ which prevents the ripping of $\bar{\mathcal{M}}_t$ for all t.

The above intricacies suggest a flowing approach to this problem that works with a qualitative resolution of the PIK $\dot{\phi}$: Core of the procedure is the choice of the sampling pair (38) or rather the 'trial' or prior part $\mathcal{T}_t$: First of all, it has to be chosen such that it allows an analytic access, or, more generally, the possibility to sample on $(\mathcal{T}_1, \bar{\mathcal{M}}_0)$ directly or via a PIK-architecture. A choice tailor made for the present task is given by

$$\mathcal{T}_t = \log \sum_{i\geq 0}^{N_t} c_{i,t}\rho_{i,t}(\phi)\,, \qquad c_{0,0} = 1\,, \quad \rho_{0,0} = \bar{\rho}_0\,, \tag{40a}$$

[**JMP:** normalisation has to be included, I'm not sure concerning the 'best' parametrisation] and Gaußian distributions

$$\rho_i(\phi) = e^{-\frac{1}{2}(\phi_n - \phi_n^{(i)})\mathcal{C}^{(i)}_{nm}(\phi_m - \phi_m^{(i)})}\,, \tag{40b}$$

with the t-dependent center $\phi^{(i)}$ of the distribution ρ_i and the t-dependent covariance $\mathcal{C}C$. In general, all these parameters are complex and vanish at $t = 0$,

$$\phi^{(i)}_{m,t}\,,\, \mathcal{C}^{(i)}_{nm,t} \in \mathbb{C}\,, \qquad \phi^{(i)}_{m,0}\,,\, \mathcal{C}^{(i)}_{nm,0} = 0\,. \tag{40c}$$

The t-dependence of the parameters in (40c) is chosen with the following constraint, or similar ones. The solution of the PIK ϕ_t with $\mathcal{T}_t = 0$ for $t < t_{\mathrm{rip}}$ with t_{rip} will

experience growing ratios

$$\left|\frac{\mathrm{Im}\left[\bar{p}_t(\phi^{(i)})\right]}{\mathrm{Re}\left[\bar{p}_t(\phi^{(i)})\right]}\right| \to \infty\,, \quad \text{for} \quad t \to t_{\mathrm{rip}}\,. \tag{41}$$

Here, t_{rip} is the flow time where the ripping occurs and $(\phi^{(i)})_t$ are the peak positions of these ratios. Importantly for us, these emerging singular points are the result of the resolution of the Wegner equation for $\dot{\phi}$ and do not have to be searched for. Now we choose $\mathcal{T}_t$ such that

$$\left|\frac{\mathrm{Im}\left[\bar{p}_t(\phi)\right]}{\mathrm{Re}\left[\bar{p}_t(\phi)\right]}\right| < M\,, \tag{42}$$

with a suitable chosen maximal value M. Not that this flow procedure provides us with the number and location $\phi^{(i)}$ of the ripping points as well as constraints for the matrix elements of the covariance, $\mathcal{C}_{mn}$. Naturally, the latter parameters are not uniquely determined while $\phi^{(i)}$ is uniquely defined as the peak position [**JMP:** Also this is not completely true as we may shift the peak position with $\mathcal{T}_t$ but who cares].

[**JMP:** Not fully checked from now on]

Importantly, such a procedure does not involve a quantitative resolution of the PIK $\dot{\phi}$ but works with a qualitative numerically cheap one as long as the ripping points are not 'fractal-type'.

After the $\mathcal{T}_t$ is determined within the above first step, we define the final sampling procedure within a second step. To that end we define the final PIK-architecture as the pair

$$(S_t\,, \qquad \mathcal{M}_0)\,, \quad S_t = \bar{S}_t - \mathcal{T}_1 = S'_t + [\mathcal{T}_t - \mathcal{T}_1]\,, \tag{43a}$$

and e.g. the real sampling pair

$$(S_0\,, \mathcal{M}_0)\,, \qquad p_0(\phi_0)d\mu_0 \in \mathbb{R}\,. \tag{43b}$$

The latter condition (43b) is one of the possible choice. Evidently, the sampling of

. (44)

B. The maximal PIKfold

Here we recall the reality condition eq.(50) in [1]. There it was formulated for PIKfolds that can be parametrised with $\sigma = \phi_R$. More generally, we have

$$\frac{d\phi_I}{d\sigma}\mathrm{Re}\left[p_t(\phi)\right] = -\frac{d\phi_R}{d\sigma}\mathrm{Im}\left[p_t(\phi)\right]\,. \tag{45}$$

Note that (45) does not exhibit the cuts which are present in the likelihood S and in the absence of cuts in p_t, (45) only exhibits poles. Bearing this in mind we may rewrite (45) in terms of the likelihood

$$\phi'_I\,\cos\left[S_{I,t}(\phi)\right] = \phi'_R \sin\left[S_{I,t}(\phi)\right]\,, \quad \phi'(\sigma) = \frac{d\phi(\sigma)}{d\sigma}\,, \tag{46}$$

with $p_t = \exp\{-S_t\}$ with $S_t = S_R + \mathrm{i}\,S_I$. Evidently, for a real action S_t, all manifolds in a PIKfold agree, $\mathcal{M}_t = \mathcal{M}_0$ (in the absence of stretching singularities). Moreover, for configurations ϕ_{sing} with $\mathrm{Re}[p_t(\phi)] = 0$, the vector field (ϕ'_R, ϕ'_I) is perpendicular to the real manifold $\{\Phi_R\} = \mathbb{R}^{\mathcal{D}}$. Note that for the choice $\sigma = \phi_R$ this leads to a ripping of the parametrisation, not necessarily the ripping of the PIKfold. However, $\dot{\phi}(\phi)$ also exhibits a singularity in this point which begs the question whether the parametrisation of $\dot{\phi}$ in terms of ϕ is a suitable one in such a case. More practically we have to sort out whether (46) can be used to defined $\mathcal{M}_1$ with $\sigma = -\infty$ with the initial condition $(\phi_R = \infty, \phi_I)$ to $\phi_R = \infty$.

ACKNOWLEDGMENTS

We thank XXX for discussions. This work is funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy EXC 2181/1 - 390900948 (the Heidelberg STRUCTURES Excellence Cluster), the Collaborative Research Centre SFB 1225 - 273811115 (ISOQUANT) and the Walter Benjamin Programme - 574031240. This work is also supported by the KAKENHI grant "Foundation of Machine Learning Physics."

Appendix A: Remnants Results

Importantly, a slightly different path choice corresponds to a different time-dependent location of the poles. For example

$$S_t(\phi) = \frac{\beta}{2U}\phi^2 - 2\log\left[1 + te^{\beta\left(i\phi+\mu+\frac{U}{2}\right)}\right]\,. \tag{A1a}$$

has time dependent poles

$$\phi_0 = (2n+1)\frac{\pi}{\beta} + i\left(\mu + \frac{U}{2} + \frac{\log(t)}{\beta}\right)\,. \tag{A1b}$$

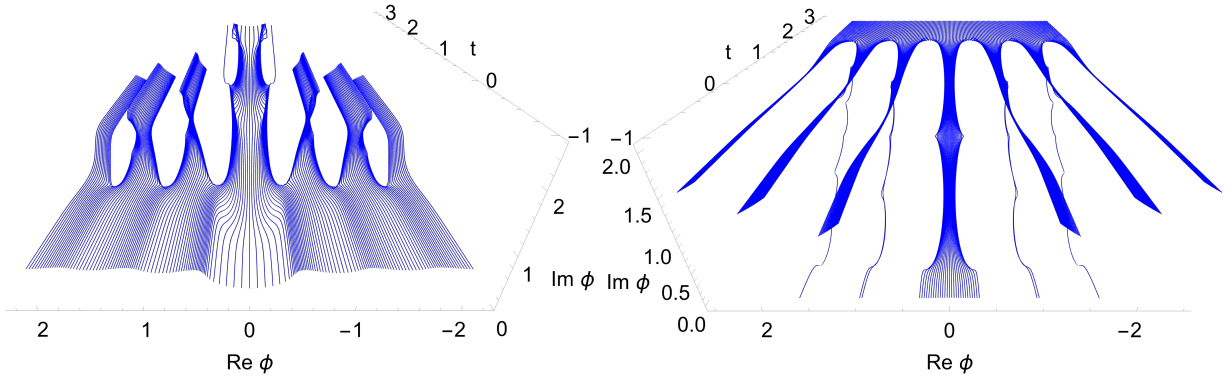
that move very strongly in the imaginary direction, starting at $-\infty$ at $t = 0$ and moving up to their final location at $t = 1$.

For the parameter choices above, we find that $\mu = 1.75$ does not yield a closed contour at $t = 1$, which is displayed in 6c. This is explained by the poles pushing though the PIKfold at some finite t.

However, we also find that the path choice Equation (27a) performs worse than (A1a) for $\mu \sim -0.5$. In these examples the initial manifold $\mathcal{M}_0 = \mathbb{R}$ lies on top of the poles (27b) of the first example, whereas the dynamics of the second path is more controlled. This is shown in Figure 6a and Figure 6b at $\mu = -0.675$ for both paths.

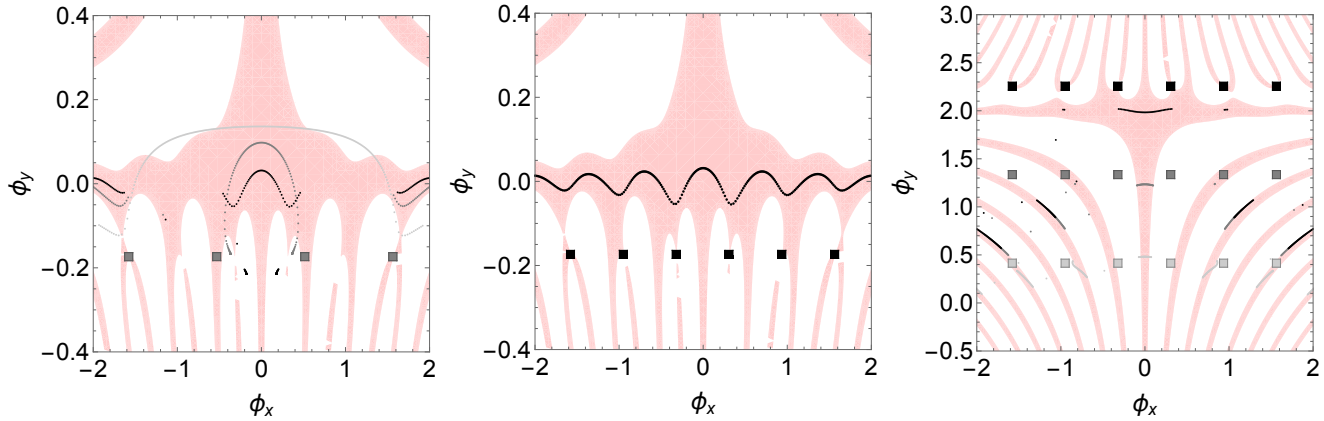
For values of $\mu < -0.7$ we find that both paths result in closed manifolds that are fit for sampling.

The conclusion is that depending on the scenario at hand the different time-dependence of the pole structure



(a) $\mu_0 = 0$ (b) $\mu_0 = 2$

FIG. 5: PIKfold.



(a) PIKfold at $t = 0.2$ (*light grey*), $t = 0.6$ (*grey*) and $t = 1$ (*black*) for the path with $\mu = -0.675$. (27a)

(b) PIKfold at $t = 1$ for the path (A1a) with $\mu = -0.675$.

(c) PIKfold at $t = 10^{-8}$ (*light grey*), $t = 10^{-4}$ (*grey*) and $t = 1$ (*black*) for the path (A1a) with $\mu = 2$.

FIG. 6: Time slices of the PIKfold for the different path and parameter choices of (27a) and (A1a) at $\beta = 10$ and $U = 1$. (b) Shows a closed manifold that can be sampled from, whereas (a, c) show a ripping of the PIKfold.

may help: It is important, that the PIKfold does not cross the poles. The fact that there are poles somewhere in the complex plane is not a problem as long as they dont intersect the manifold throughout the time evolution. This also highlights the very "local" structure of the PIKs.

Appendix B: PIKs & Lefschetz thimbles approach

In this section we discuss parallels and crucial differences between the complex PIKs and the Lefschetz thimble approach. By applying the *target action* strategy to computing thimbles, we find a semi-analytic strategy to compute the thimble solutions and to sample from them.

We begin with a brief introduction to Lefschetz-thimbles in Section B 1. This is followed by the construction of Lefschetz-thimbles from a complex PIK in Section B 2.

1. A brief repetition of Lefschetz-thimbles

Similar to the complex PIK architecture, Lefschetz thimbles (LT) use the strategy of contour deformation to attack the sign problem. In distinction to the former, the LT-contour is made up of the so-called thimbles $\mathcal{J}_\sigma$ on which the phase of the integrand is constant. The thimbles are computed as follows: Firstly, every thimble is attached to a critical point ϕ_σ of the action given by

$$\frac{\partial S}{\partial \phi}(\phi_\sigma) = 0\,. \tag{B1}$$

Then, the thimble $\mathcal{J}_\sigma$ is then defined as the set of points obtained from the downward gradient flow

$$\frac{d\phi}{dt} = \overline{\left(\frac{\partial S}{\partial \phi}\right)}\,, \tag{B2}$$

with the boundary condition $\phi_{t\to-\infty} = \phi_\sigma$. By considering

$$\left.\frac{d}{dt}S(\phi)\right|_{\mathcal{J}_\sigma} = \frac{\partial S}{\partial\phi}\overline{\left(\frac{\partial S}{\partial\phi}\right)} = \left|\frac{\partial S}{\partial\phi}\right|^2 , \tag{B3}$$

we find that along (B2), the real part of the action increases monotonically, while its imaginary part remains constant. The former property implies that the critical points are the leading contribution to the sampling procedure.

The sampling contour is a linear combination of all thimbles,

$$\mathcal{C} = \sum_\sigma n_\sigma, \mathcal{J}_\sigma , \tag{B4}$$

where the coefficients $n_\sigma \in \mathbb{Z}$ are intersection numbers of the corresponding anti-thimbles with the real axis. The anti-thimbles are the result of the flow (B2) with the boundary condition $\phi_{t\to+\infty} = \phi_\sigma$. Finally, the partition function can be written as a sum of integrals over the contributing thimbles

$$Z = \sum_\sigma n_\sigma \, e^{-i\,\mathrm{Im}S(\phi_\sigma)} \int_{\mathcal{J}_\sigma} d\mu_\sigma \, e^{-\mathrm{Re}S(\phi)} . \tag{B5}$$

In general $\mathcal{J}_\sigma$ is a curved manifold/contour in a complex space $\mathbb{C}^{\mathcal{D}}$, with a complex measure $d\mu_\sigma$. Hence the integrals over single thimbles $\mathcal{J}_\sigma$ have to be rewritten in terms of a real parametrisation. For the single-site example we can choose $\phi(s)$ with $s \in \mathbb{R}$. Then the integral over a single thimble reads

$$Z_\sigma = \int ds \det\left(\frac{\partial\phi}{\partial s}\right) e^{-S(\phi(s))} . \tag{B6}$$

The constant phase factor in the action can be factored out, however the Jacobi-determinant $\det\left(\frac{\partial\phi}{\partial s}\right)$ is not real in general, which leaves one with a residual phase

$$\theta(s) = \arg\det\left(\frac{\partial\phi}{\partial s}\right) - \mathrm{Im}\, S(\phi_\sigma) . \tag{B7}$$

Expectation values of observables are then computed by re-weighting with the residual phase

$$\langle\mathcal{O}\rangle = \frac{\int ds\, P(s)\, e^{i\theta(s)}\mathcal{O}(\phi(s))}{\int ds\, P(s)\, e^{i\theta(s)}} , \tag{B8a}$$

with the real weight

$$P(s) = \left|\det\left(\frac{\partial\phi}{\partial s}\right)\right| e^{-\mathrm{Re}\, S(\phi(s))} . \tag{B8b}$$

In conclusion, the LT approach uses a decomposition of the contour into different thimbles. While the phase on a specific thimble is constant and sampling can be performed after a re-weighting, the phases of two different thimbles are generically different, which can lead to large cancellations and a residual sign problem. Moreover, for many systems the total number of thimbles is very large, reintroducing the sign problem through the back door.

2. Thimbles through the PIK-lens

The complex PIK architecture builds on the Wegner equation which considers general reparametrisations of the theory. As such, it is very flexible in its choice of the initial sampling manifold, and the path with which we deform the action. For the aim of the present section, i.e. discussing the embedding of the Lefschetz thimble approach in the PIK, we restrict ourselves to real initial sampling manifolds

$$(p_0(\phi)\,, \mathcal{M}_0)\ , \qquad \mathcal{M}_0 = \mathbb{R}\,. \tag{B9}$$

This choice is made for simplicity, and allows to use the initial real variable $\phi_0 \in \mathbb{R}$ as parametrisation for the complex sampling manifold or complex integration contour $\phi_1(\phi_0) \in \mathcal{M}_1$.

As already noted previously, the thimble approach and the complex PIK follow a similar strategy by deforming the contour. However, we find that there are two key differences:

(a) There is only one final sampling manifold for the complex PIK, in distinction to the thimble approach, where there can be as many thimbles as there are critical points (B1).

(b) The thimble aims for a constant phase on a curved manifold, which leaves one with the residual phase (B7) containing the Jacobi-determinant of the real parametrisation. In contrast, the parametrisation of the complex contour is already part of the weight-preserving property of the complex PIK, effectively achieving $\theta(\phi_0) = 0$ on the entire PIKfold $\mathcal{M}_t$.

To illustrate (b), we recall the weight preserving property of the PIK, which reads

$$\begin{aligned}\int_{\mathcal{M}_t} d\mu_t \, e^{-S_t(\phi_t)} &= \int_{\mathcal{M}_0} d\mu_0 \det\left[\frac{\partial\phi_t}{\partial\phi_0}\right] e^{-S_t\left(\phi_t(\phi_0)\right)} \\ &\equiv \int_{\mathcal{M}_0} d\mu_0 \, e^{-S_0(\phi_0)} ,\end{aligned} \tag{B10}$$

and compare it to (B8): The important difference is, that following the second equality, the product of the Jacobi-determinant and the exponential of the action have a constant phase. Even better, they are completely real, whereas the LT constructs only an action with a constant phase but leaves the parametrisation of the manifold out. The inclusion of the Jacobian into the definition of the PIK is the crucial difference between both approaches. In fact, in systems where $\det\left[\frac{\partial\phi_t}{\partial\phi_0}\right] = \text{const.}$ the complex PIK and the LT describe the same manifold.

a. Constructing Lefschetz-thimbles from the complex PIK

In this section we discuss how LT's can be computed within the complex PIK framework. To this aim, we go back to the more general considerations about the *physics-informed* pair (PIRG-pair)

$$\Big(S_t(\phi), \dot\phi(\phi)\Big)\,. \tag{B11}$$

For the complex PIK applications in [1] we have defined $S_t(\phi)$ with a unique path and proceeded with computing the resulting $\dot\phi$ from the Wegner equation (C4). However, as seen in applications within the fRG [4**? ? ?**] it can make sense to use the reparametrisation freedom in the Wegner equation differently, i.e. by implementing a different, additional constraint instead of completely fixing the path of the action.

In the present case, we want to use this freedom to implement a generative architecture that samples on the Lefschetz thimble. To this aim we define a path $\tilde S_t$ for the action, which starts with an easy distribution on the real axis and evolves towards our target distribution on the thimble

$$\Big(e^{-\tilde S_0(\phi)}\,,\,\mathbb{R}^{\mathcal D}\Big) \longrightarrow \Big(e^{-\tilde S_1(\phi)}\,,\,\mathcal{J}_\sigma\Big)\,. \tag{B12}$$

Note our choice of notation $\tilde S_t$: This path does not correspond to the action in the PIRG-pair, it is simply a tool to construct a kernel $\dot\phi$ which fulfils

$$0 \overset{!}{=} \partial_t \tilde S_t(\phi_t) = \partial_t \tilde S_t(\phi_t) + \partial_t\phi_t\,\partial_{\phi_t}\tilde S_t(\phi_t)\,, \quad \dot\phi := \partial_t\phi_t\,. \tag{B13}$$

Since the initial $\tilde S_0$ is real, (B13) ensures that all $\tilde S_t$ and importantly $\tilde S_1$ on the thimble are also real. Note that this allows for a complex phase, as

$$\tilde S_t = \log\Big(\tilde{\mathcal{N}}_t\Big) + \tilde{\tilde S}_t\,, \tag{B14}$$

contains a normalisation term.

The normalisation term in (B14) is uniquely determined by requiring that the thimble remains attached to an evolving critical point (B1)

$$\dot\phi(\phi_{\sigma,t}, \tilde{\mathcal{N}}_t) \overset{!}{=} \partial_t\phi_{\sigma,t} = -\frac{\partial_t \tilde S_t^{(1)}(\phi_{\sigma,t})}{\tilde S_t^{(2)}(\phi_{\sigma,t})}\,, \tag{B15}$$

where the second equality is derived by requiring that $\partial_t \tilde S^{(1)}(\phi)|_{\phi=\phi_\sigma} = 0$, which simply says that the critical point is defined by its vanishing first derivative at all times.

Together (B13) and (B14) yield a simple algebraic equation to compute the kernel. If we restrict ourselves to paths $\tilde S_t$ where the total number of critical points is preserved, each initial critical point allows to compute a different final thimble.

In distinction to the standard complex PIK procedure, (B13) does not allow for straightforward sampling. In fact, evaluating the operators directly using (13), simply evaluates the distribution (B6) containing the Jacobian of the curved contour. In this sense, the kernel defined in (B13) describes a weight-preserving complex PIKfold for the target distribution

$$S_t(\phi) = \tilde S_t(\phi) + \operatorname{Tr}\log\left[\frac{\partial\phi_t}{\partial\phi_0}(\phi_0)\right]\Big|_{\phi_0=\phi_0(\phi_t)}\,, \tag{B16}$$

which one can also compute by solving the Wegner equation (C4) using the kernel (B13). This computation is of course much more difficult, as we have to solve a partial differential equation.

To obtain observables of the action $\tilde S_t(\phi)$ one can simply resort so re-weighting in the standard way (B8). For the generated samples this corresponds to summing

$$\langle\mathcal{O}(\phi)\rangle_{\tilde p_t} = \frac{\sum_n \mathcal{O}(\phi_t(\phi_0))\det\left[\frac{\partial\phi_t}{\partial\phi_0}(\phi_0)\right]}{\sum_n \det\left[\frac{\partial\phi_t}{\partial\phi_0}(\phi_0)\right]}\,. \tag{B17}$$

Note, that the quality of results is expected to be worse for steeper slopes in the Jacobian. On the plus-side we have semi-analytics access to the thimble.

Appendix C: Remnants

1. Complex sampling and the sign problem

PIK-architectures [2] have been mainly developed for the efficient sampling with normalised distributions of lattice field theories. In the present work we consider a scalar field theory with a real scalar field and a generic complex distribution for illustrative purposes. We define

$$p(\varphi) = e^{-S(\varphi)} \in \mathbb{C}\,, \qquad S(\varphi) = \hat S(\varphi) + \log\mathcal{N}\,. \tag{C1a}$$

Here, $S(\varphi) \in \mathbb{C}$ is the action of the complex distribution $p(\varphi) \in \mathbb{C}$, with $\hat S(\varphi = 0) = 0$. The constant term $\log\mathcal{N} \in \mathbb{C}$ is nothing but the normalisation of the distribution $p(\varphi)$ which guarantees

$$Z = \int_{\mathcal M} d\varphi\, p(\varphi) = 1\,, \qquad \int_{\mathcal M} d\varphi = \prod_i \int_{\mathbb{R}} d\varphi_i\,. \tag{C1b}$$

where i labels the data points. In our lattice example, the *real* field φ is defined on a finite (hyper-)cubic lattice $\mathcal{D}$ with

$$\varphi : \mathcal{D} \to \mathcal{M} = \mathbb{R}^{\mathcal D}\,, \qquad \mathcal{D} = [0, 1, ..., N_{\max}]^d\,. \tag{C1c}$$

We will not specify the action $\hat S(\varphi)$; typically it includes a kinetic term (hopping term) and an on-site self-interaction (potential term). For such an example of a real scalar lattice field theory we refer the reader to [2]. In the general setup, $\mathcal{D}$ labels a continuous or discrete data

set with data φ, and the continuous or discrete sampling manifold $\mathcal{M}$ is generalised likewise. All these setups are defined by *sampling pairs* $(p(\varphi)\,,\mathcal{M})$ of the probability distribution and the sampling manifold.

In general, sampling with a complex distribution $p(\varphi)\in\mathbb{C}$ in (C1a) poses a numerical challenge, and for highly oscillatory distributions the sampling task may exhibit a sign problem. A rather generic and seemingly simple resolution of such a sign problem in our example (C1) is provided by the deformation of the integration manifold $\mathcal{M}$ with a highly oscillatory distribution $p(\varphi)$ into an integration manifold $\mathcal{M}_1\subset\mathbb{C}^{\mathcal{D}}$, on which $p(\varphi)$ exhibits no sign problem.

Assuming that this deformation step has been performed, it leaves us with a secondary problem of efficiently sampling from the given distribution $p(\varphi)$. This secondary problem can be remedied by mapping $p(\varphi)$ to a simple distribution such as a normal distribution $p_0(\phi)$ where $\phi\in\mathcal{M}_0\subset\mathbb{C}^{\mathcal{D}}$. While normal distributions with the sampling manifold $\mathcal{M}_0=\mathcal{M}$ are standard choices, the set of viable $(p_0,\mathcal{M}_0)$ is not restricted to them. The only requirement is efficient sampling while keeping all symmetries.

2. Complex sampling with physics-informed kernels

In the present work we use the PIK-architecture [2] to resolve the combined sign problem and efficient sampling tasks discussed above: in general, the physics-informed kernel (PIK) provides a map from some sampling pair $(p_0(\phi)\,,\mathcal{M}_0)$ to that under investigation,

$$(p_0(\phi)\,,\mathcal{M}_0)\longrightarrow(p_1(\phi)\,,\mathcal{M}_1)\,,\qquad p_1(\phi)=p(\phi)\,. \tag{C2}$$

Consequently, the PIK-architecture allows for an efficient sampling of the pair $(p(\phi)\,,\mathcal{M}_1)$, if the original pair can be sampled efficiently. This solves the given sampling task if the sampling on $\mathcal{M}_1$ is equivalent to that on the original sampling manifold $\mathcal{M}$,

$$\int_{\mathcal{M}_1} d\mu_1\, p(\phi)\,\mathcal{O}(\phi)\stackrel{!}{=}\int_{\mathcal{M}} d\mu\, p(\varphi)\,\mathcal{O}(\varphi)\,, \tag{C3}$$

for general operators $\mathcal{O}(\phi)$. Here, $d\mu_1(\phi)$ and $d\mu(\varphi)$ are the volume elements of $\mathcal{M}_1$ and $\mathcal{M}$, embedded in the complexification $\mathcal{C}$ of $\mathcal{M}$ and φ denotes the variable in the original complex sampling problem. Equation (C3) is satisfied if $\mathcal{M}_1$ can smoothly be deformed into $\mathcal{M}$ without swiping over singular points in the distribution $p(\phi)$ and the operators $\mathcal{O}(\phi)$.

In our lattice example the sampling manifold is given by $\mathcal{M}=\mathbb{R}^{\mathcal{D}}$ and its complexification is given by $\mathcal{C}=\mathbb{C}^{\mathcal{D}}$. For the sake of simplicity we will consider this explicit example from now on without loss of generality.

The key property of the PIK-map in (C2) is that it is weight-preserving. This entails in particular, that the sampling with $(p_1(\phi)\,,\mathcal{M}_1)$ is real, *if* the sampling pair $(p_0(\phi)\,,\mathcal{M}_0)$ is real. More generally, if the, potentially complex, sampling pair $(p_0(\phi)\,,\mathcal{M}_0)$ can be sampled efficiently and has no sign problem, it induces an efficient and sign- and overlap-problem free sampling for the sampling pair $(p_1(\phi)\,,\mathcal{M}_1)$.

We proceed by discussing the computation of the physics-informed kernel (PIK), that is the differential map $\dot\phi(\phi)$. In the PIK-architecture [2] the local deformation of the sampling manifold and the distribution is governed by the Wegner equation [3],

$$\frac{d\,p_t(\phi)}{dt}=-\frac{\partial}{\partial\phi_i}\Big[\dot\phi_{t,i}(\phi)\,p_t(\phi)\Big]\,,\qquad t\in[0,1]\,, \tag{C4}$$

with the corresponding PIK $\dot\phi_t(\phi)$ of the deformation, and the RG-time (renormalisation group time) t. The index $i\in\mathcal{D}$ in (C4) is summed over the data points. Equation (C4) depends on a t-dependent normalised distribution

$$p_t(\phi)=e^{-S_t(\phi)}\,, \tag{C5}$$

with $S_t=\hat S_t+\log\mathcal{N}_t$. Importantly, $\hat S_t$ is chosen analytically to interpolate between $p_0(\phi)$ and $p_1(\phi)$. In our lattice example ϕ is the complex field variable on the lattice $\mathcal{D}$,

$$\phi:\mathcal{D}\to\mathbb{C}^{\mathcal{D}}\,. \tag{C6}$$

In general, $\mathcal{D}$ is the set of data points and ϕ takes values in the complexification of the data range.

The complexification of ϕ has two sources. The first rather trivial one is the possible choice of a complex initial manifold $\mathcal{M}_0\subset\mathbb{C}^{\mathcal{D}}$. The second source is the deformation $\partial_t\mathcal{M}_t$ of the manifold $\mathcal{M}_t$ due to the complex transformation $\dot\phi_t$ itself. With (C5), we rewrite (C4) as a differential equation for the physics-informed RG-pair (PIRG pair) $(S_t(\phi)\,,\dot\phi)$,

$$\frac{d\,S_t(\phi)}{dt}+\dot\phi_t(\phi)\,\frac{\partial S_t(\phi)}{\partial\phi}=\frac{\partial\dot\phi_t(\phi)}{\partial\phi}\,, \tag{C7}$$

where we have suppressed the indices $i\in\mathcal{D}$ for the sake of readability. For complex $dS_t(\phi)/dt$ and/or $\partial S_t(\phi)/\partial\phi$, the map $\dot\phi(\phi)$ is complex and so is the differential manifold $\partial_t\mathcal{M}_t$. Note that for smooth $\dot\phi(\phi)$, the manifold $\partial_t\mathcal{M}_t\subset\mathbb{C}^{\mathcal{D}}$ is a closed or open hyper-surface, an example with one complex dimension and further explanations can be found in the supplement.

Accordingly, (C4) and (C7) comprise a general differential reparametrisation of the path integral with a smooth deformation of the integration manifold with the PIK $\dot\phi(\phi)\in\mathbb{C}^{\mathcal{D}}$, if the latter is invertible and non-singular.

Now we set up the complex PIK-architecture for the sign-problem free sampling task: With (C4) and (C7) we map the sampling with $p(\varphi)$ on $\mathcal{M}$ with a sign problem

to the efficient sampling with $p_0(\phi)$ on the sign-problem free manifold $\mathcal{M}_0$. We find for all t,

$$\int_{\mathcal{M}_t} d\mu_t\, p_t(\phi)\, \mathcal{O}(\phi) = \int_{\mathcal{M}_0} d\mu_0\, p_0(\phi)\, \mathcal{O}\big(\phi_t(\phi)\big)\,, \qquad \text{(C8a)}$$

where

$$\phi_t(\phi) = \phi + \int_0^t ds\, \dot\phi_s(\phi_s)\,, \qquad \varphi(\phi) = \phi_1(\phi)\,, \qquad \text{(C8b)}$$

and $d\mu_t$ denotes the volume element of the corresponding manifold $\mathcal{M}_t$. Equation (C8) with (C3), evaluated for $t = 1$, leads to the aimed for equivalence

$$\int_{\mathcal{M}} d\mu\, p(\varphi)\, \mathcal{O}(\varphi) = \int_{\mathcal{M}_0} d\mu_0\, p_0(\phi)\, \mathcal{O}\big(\varphi(\phi)\big)\,. \qquad \text{(C9)}$$

The existence of the global map (C8b) was discussed in detail in [2] for real actions S_t and the existence criteria apply to complex S_t without modification. The proof of (C9) as well as further details can be found in the supplement. Here we only emphasise that the physics or more generally the complete information of the underlying distribution $p(\varphi)$ is carried by the map $\varphi(\phi)$ alone. The analytic access to this map, and the factorisation of the global computing task into the independent ones of computing $\dot\phi_t(\phi)$ for all $t \in [0,1]$ are amongst the advantageous properties of the PIK-architecture, see again [2, 4].

It is left to prove that the right-hand side of (C9) is sign-problem free. For that purpose we show, that the sampling with $p_t(\phi)$ on $\mathcal{M}_t$ has no sign and overlap problems, if the sampling with $(p_0(\phi)\,, \mathcal{M}_0)$ has no sign problem. This important property can be deduced from the *weight-preserving* property of the PIK-architecture. For that purpose we shall prove that the flow of the 'statistical' weight of general intervals $\mathcal{T}_t$ vanishes, that is

$$\frac{d\, P_t(\mathcal{T}_t)}{dt} = 0\,, \qquad P_t(\mathcal{T}_t) = \int_{\mathcal{T}_t} d\mu_t\, p_t(\phi)\,. \qquad \text{(C10)}$$

Here, $\mathcal{T}_t$ is the image of a given interval $\mathcal{T}_0$ with the map $\phi_t(\phi)$ in (C8b). A direct proof of (C10) with (C4) is provided in the supplement. Here, we simply consider the integral of (C10) from the initial RG-time to t, which is $P_t(\mathcal{T}_t) = P_0(\mathcal{T}_0)$. This identity can be derived from (C8) by using the characteristic function $\chi_{\mathcal{T}_t}$ of the sub-manifold $\mathcal{T}_t$ as the operator $\mathcal{O}$. Then, $\chi_{\mathcal{T}_t}(\phi_t(\phi)) = \chi_{\mathcal{T}_0}(\phi)$ is the characteristic function of $\mathcal{T}_0$ on $\mathcal{M}_0$. Hence, inserting this operator in (C8) leads us to

$$\int_{\mathcal{T}_t} d\mu_t\, p_t(\phi) = \int_{\mathcal{T}_0} d\mu_0\, p_0(\phi)\,, \qquad \text{(C11)}$$

and the t-derivative of (C11) leads us to (C10). If we use infinitesimal sub-manifolds $\mathcal{T}_t$, we readily find that the differential distribution or weight $d\mu_t\, p_t(\phi)$ is equivalent to $d\mu_0\, p_0(\phi)$. The differential equivalence is elucidated with the pull-back of the left-hand side to the pre-image $\mathcal{T}_0$, to wit,

$$\int_{\mathcal{T}_0} d\mu_0 \det\left[\frac{\partial \phi_t}{\partial \phi_0}\right] e^{-S_t\left(\phi_t(\phi_0)\right)} \equiv \int_{\mathcal{T}_0} d\mu_0\, e^{-S_0(\phi_0)}\,. \qquad \text{(C12)}$$

As (C12) holds true for general sub-manifolds $\mathcal{T}_0$, it has to hold true for the integrands, and we arrive at

$$S_t\big(\phi_t(\phi_0)\big) - \operatorname{Tr}\log\left[\frac{\partial \phi_t}{\partial \phi_0}(\phi_0)\right] = S_0(\phi_0)\,. \qquad \text{(C13)}$$

In short, (C10) or (C11) entail that the map $\dot\phi$ is weight-preserving by construction and hence no overlap problem and sign problem is generated by the flow with $\dot\phi$. Hence, if the sampling pair at $t = 0$ is chosen sign-problem free, so are all sampling pairs. In particular, if the sampling pair at $t = 0$ is real, all sampling pairs are real, and specifically $(p_1(\phi)\,, \mathcal{M}_1)$. We call the set of weight-preserving manifolds

$$\mathcal{P} = \mathcal{P}_1\,, \qquad \mathcal{P}_t = \{\mathcal{M}_s \,|\, s \in [0,t]\}\,, \qquad \text{(C14)}$$

endowed with the holomorphic map $\varphi(\phi)$ on $\mathcal{P}$, the *PIKfold* of the set of flowing distributions $\{p_t\}$. The sets $\mathcal{P}_t$ are sub-PIKfolds. More details are provided in the supplement. Sign-problem free PIK-architectures are obtained by construction for a sign-problem free sampling pair $(p_0, \mathcal{M}_0)$.

[**JMP:** end complex PIK paper [1], new from now on]

[1] F. Ihssen, R. Kapust, and J. M. Pawlowski, (2026), arXiv:2603.03159 [hep-lat].

[2] F. Ihssen, R. Kapust, and J. M. Pawlowski, (2025), arXiv:2510.26678 [hep-lat].

[3] F. J. Wegner, Journal of Physics C: Solid State Physics **7**, 2098 (1974).

[4] F. Ihssen and J. M. Pawlowski, Annals Phys. **481**, 170177 (2025), arXiv:2409.13679 [hep-th].